



SIMATS SCHOOL OF ENGINEERING SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES CHENNAI-602105



Team Members:

Enamala Gowtham (192311190)

A. Ganesh Ram Kamal (192311183)

G. Abhinay (192373062)

Course Details:

Course: Internet Programming

Code: CSA4305

Faculty: Dr D Sudhagar

Web-Based Student Skill Assessment and Development Management System

One-Page Individual Reflection

Working on the Student Skill Development Management System gave me an opportunity to apply the concepts learned in Internet Programming to a practical project. The main purpose of our system is to help students manage their skills, assessment results, certifications, projects, and development progress in one place. Before starting the development, we discussed the required features and divided the system into different modules. This made the development process easier and helped us work on each part in an organized manner.

For the design, we decided to keep the interface simple and easy to use. HTML was used to create the structure of the web pages and forms. CSS was used to design the layout and improve the appearance of the application. JavaScript was used to implement interactive functions, including form validation, skill management, assessment processing, and skill-gap calculation. We also used XML and XSLT to demonstrate the concepts covered in the course. The prototype uses browser storage to maintain the information entered during execution.

During development, we faced some challenges while connecting the different modules and checking whether the entered information was processed correctly. We also had to test the skill-gap calculation with different values. For example, when the required skill level was higher than the current level, the system had to identify that improvement was needed. We tested different inputs, corrected errors, and repeated the testing process until the main functions worked as expected.

This project improved my understanding of HTML, CSS, and JavaScript. I learned how these technologies can be combined to create an interactive web application. I also gained practical knowledge of XML and XSLT and understood how structured information can be represented and displayed. In addition to technical skills, I improved my debugging and problem-solving abilities. Working with the team also helped me understand the importance of dividing tasks and coordinating individual work.

The project is relevant to SDG 4 – Quality Education because it helps students identify their current skills and areas that require improvement. It is also connected to SDG 8 – Decent Work and Economic Growth because students can maintain their skills, certifications, and project experience for future employment. The project also supports SDG 9 – Industry, Innovation and Infrastructure by using web technologies to create a digital platform for skill development. These SDGs are identified in the assignment specification.

The assignment contributed to the mapped Course Outcomes by giving me practical experience in web-page development and dynamic web programming. I applied CSS concepts while designing the interface and JavaScript concepts while developing interactive features. I also gained experience with XML and XSLT as part of the web application. The assignment therefore helped me connect the concepts learned in class with their practical implementation.

Overall, this assignment improved my confidence in developing web applications. It taught me that successful development requires proper planning, coding, testing, debugging, and teamwork. The experience also helped me understand how Internet Programming concepts can be used to develop a useful solution for student skill development.